Chen Xu

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EDUCATION

New York University **Expected May 2029**

B.S. in Computer Science **GPA:** 4.0

- Core Member of the ICPC Team (Poly Programming Club)
- Member of the Tennis Club

COMPETITIVE PROGRAMMING

- Codeforces International Master; USACO Platinum; CSP-J National Champion (1st nationally)
- **ICPC Regionals**: 11th of 66 teams at ICPC Greater New York Regional, 2026.
- **ICPC Gold Medal:** 4th of 237 teams at ICPC/CCPC Shandong Provincial Contest, 2026.
- **ICPC Gold Medal:** 38th of 418 teams at 50th ICPC Asia Xi'an Regional, 2025.
- **ICPC Silver Medal:** 82nd of 280 official teams at 49th ICPC Asia East Continent Final, 2024.
- **CCPC Silver Medal:** 60th of 324 teams at 11th CCPC Chongqing Regional, 2025.

EXPERIENCE

New York University **Nov 2026 – Present**

Undergraduate Researcher, advised by Prof. Torsten Suel

- Optimizing a C++ reimplementation of SPRAWL (SIGIR '25), an early-termination algorithm for top-k search retrieval
- Built a packed two-pass radix accumulator that merges candidate scores 6–9× faster than the hash and sort versions
- Cut average query latency 32–37% with identical top-k results on 6,980 queries across BM25 and DeepImpact indexes

Dacheng Precision Equipment **Feb – Aug 2026**

Computer Vision Engineer

- Lead developer of a CCD system measuring battery-electrode widths on 30+ production machines at 5 manufacturers
- Implemented chessboard camera calibration for the measurement pipeline
- Wrote an OpenCV/RANSAC edge detector that ignores foil wrinkles and scratches, matching ground truth at $R^2 > 0.995$
- Built the QC step that checks each measured electrode width against tolerance
- Integrated with the inspection line's control system over real-time TCP; shipped 50+ releases from plant feedback

PROJECTS

Usage-Aware GPU Memory Oversubscription | HAMi (open-source Kubernetes GPU virtualization) **2025**

- Found HAMi oversubscribes GPU memory by requested quota, ignoring real usage, so co-located jobs hit CUDA OOM
- Changed its Go scheduler to admit pods only when real usage plus the new request fits in physical memory
- Raised GPU utilization from 55% to 78% over a tuned baseline on a real Kubernetes V100 cluster

BookaCook | iOS Application ($20K Hackathon Grand Prize, 100+ downloads) **2025**

- Shipped full-stack iOS recipe platform; built Python web crawlers to scrape 1,200+ cookbooks and 1M+ recipes
- Built a pipeline that turns TikTok videos into structured recipes using Whisper, GPT-4o and Llama 3

SKILLS

Languages: C++, Python, Go, Swift

Technologies: PyTorch, OpenCV, Kubernetes, Git